

Software Test Plan (STP) - Traffic Management System (Simulation)

Project: Traffic Management System (Simulation)

Version: 1.0

Authors: Team Number: Team 8

Instructor: Dr. Pradeep Kumar

Team Members:

- PES2UG24CS131 — Chetan
- PES2UG24CS159 — Dimpal N
- PES2UG24CS176 — Halahally Shivaprasad Deeksha Prasad
- PES2UG24CS816 — Nikitha V

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1. Introduction

Purpose

This document defines objectives, scope, strategy, resources, schedule, responsibilities, risks, traceability, and acceptance criteria for testing the Traffic Management System.

Scope

Testing covers vehicle addition/processing, queues, multiple lanes, signals, emergency priority, density, statistics, reset, validation, exit, non-functional behavior, and security. Real-world traffic hardware and infrastructure are excluded.

References

Traffic Management System SRS v1.0; Traffic Management System SAD v1.0; project source and automated tests.

Definitions

TMS, FR, NFR, SR, RTM, TC.

2. Test Items

- Vehicle/Queue module
- Multiple Lane Management
- Traffic Signal
- Emergency Priority
- Traffic Monitoring
- Statistics
- Simulation Control
- Input Validation
- CLI/User Interface
- Security/Robustness

3. Features to be Tested

SRS IDs	Feature
TMS-F-001-003	Vehicle addition, processing, queue display
TMS-F-004-006	Multiple lanes, selection, empty/full queues
TMS-F-007-009	Signal and emergency priority
TMS-F-010-012	Density, queue monitoring, statistics
TMS-F-013-015	Reset, validation, exit
TMS-NF-001-005	Performance, reliability, robustness, usability, maintainability
TMS-SR-001-005	Security validation

4. Features Not to be Tested

- Real-world traffic signal hardware
- Roadside sensors/vehicle hardware
- External traffic infrastructure
- Network/cloud backends
- Database persistence, since current design is in-memory

5. Test Approach / Strategy

Levels

- Unit testing — queue, validation, priority, signal, monitoring, statistics
- Integration testing — module interactions
- System testing — end-to-end CLI workflows
- Acceptance testing — SRS acceptance criteria

Types

- Functional
- Boundary/negative
- Regression

- Performance
- Usability
- Security/robustness
- Maintainability/static checks

Entry Criteria

Build compiles; environment and test data are ready; SRS and test cases are baselined.

Exit Criteria

100% planned cases executed; all high-priority criteria verified; 0 critical defects open; complete requirement coverage.

5.1 Security Validation

- Validate lane bounds.
- Validate queue capacity.
- Test invalid vehicle IDs/input.
- Confirm input cannot execute OS commands.
- Confirm no confidential data is stored/displayed.

6. Test Environment

Area	Environment
Hardware	Standard development computer
Software	C compiler such as GCC; terminal; executable
Repository	Git/GitHub
Tools	Compiler/test runner; GitHub Actions if enabled; PlantUML/draw.io
Test Data	Normal/emergency vehicles, valid/invalid lanes, empty/full queues, boundary input

7. Test Schedule

Milestones:

- Test case design: 05-Sep-2025
- Environment setup: 07-Sep-2025
- Test execution start: 08-Sep-2025

- Test execution end: 20-Sep-2025
- UAT: 22-Sep-2025 to 25-Sep-2025

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Chetan	Prepare plan, coordinate execution
Test Engineer	Dimpal N	Design & execute test cases, log defects
Developer	Halahally Shivaprasad Deeksha Prasad	Support defect fixes and triage
Product Owner	Nikitha V	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Build instability	Compile and smoke-test before execution
Queue boundary defects	Dedicated empty/full tests
Priority defect	Test normal and emergency vehicles together
Incomplete coverage	Maintain RTM
Environment differences	Document compiler/environment

11. Assumptions & Dependencies

- Project builds with a standard C compiler.
- SRS is the behavior baseline.

- Testers have source/executable access.
- No external services are required.
- Unspecified implementation details will be verified against finalized source/design.

12. Suspension & Resumption Criteria

Suspend testing if the application cannot compile/start, a blocking defect prevents more than 30% of cases, or the environment is unavailable. Resume when blocking defects are resolved, a stable build is available, and the environment/test data are restored.

13. Test Case Management & Traceability

Every SRS requirement is mapped to at least one test case.

Requirement ID	Primary Test Case
TMS-F-001	TC-TMS-001
TMS-F-002	TC-TMS-002
TMS-F-003	TC-TMS-003
TMS-F-004	TC-TMS-004
TMS-F-005	TC-TMS-005
TMS-F-006	TC-TMS-006
TMS-F-007	TC-TMS-007
TMS-F-008	TC-TMS-008
TMS-F-009	TC-TMS-009
TMS-F-010	TC-TMS-010
TMS-F-011	TC-TMS-011
TMS-F-012	TC-TMS-012
TMS-F-013	TC-TMS-013
TMS-F-014	TC-TMS-014
TMS-F-015	TC-TMS-015
TMS-NF-001	TC-NF-001

TMS-NF-002	TC-NF-002
TMS-NF-003	TC-NF-003
TMS-NF-004	TC-NF-004
TMS-NF-005	TC-NF-005
TMS-SR-001	TC-SEC-001
TMS-SR-002	TC-SEC-002
TMS-SR-003	TC-SEC-003
TMS-SR-004	TC-SEC-004
TMS-SR-005	TC-SEC-005

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Chetan	
Dev Lead	Halahally Shivaprasad Deeksha Prasad	
Product Owner	Nikitha V	
Test Engineer	Dimpal N	